

# NOTES ON SIZE, SEX RATIO, AND SPAWNING OF THE SPINY LOBSTER *PANULIRUS GUTTATUS* (LATREILLE), NEAR MIAMI BEACH, FLORIDA<sup>1</sup>

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## ABSTRACT

Spiny lobsters, *Panulirus guttatus* (Latreille), were collected at night along the jetties bordering the ship channel near Miami Beach, Florida. Observations on size distribution, sex ratio, and proportion of ovigerous females in the catch of lobsters during June through October 1970, are presented.

## INTRODUCTION

The spiny lobster, *Panulirus guttatus* (Latreille), is occasionally taken in the commercial catches from eastern Florida (Smith, 1954, 1958), but it is less abundant than *P. argus* (Boone, 1930). Moore (1962) considered *P. guttatus* to be rare in comparison to *P. argus* at Palm Beach, Florida. Though *P. guttatus* appeared to be present during all seasons of the year, it was seldom captured (*op. cit.*). Sims (1966) reported that *P. guttatus* has at times been captured in abundance in the area of West Palm Beach, Florida.

Sutcliffe (1953) made observations on size distribution, sex ratio, and proportion of ovigerous females among 158 specimens of *P. guttatus* collected in Bermuda. We made similar observations on *P. guttatus* collected in Government Cut (Fig. 1), the ship channel passing between Miami Beach and Fisher Island, Florida, during June through October 1970.

## AREA AND METHODS

The jetties bordering Government Cut consist of large granite and limestone boulders. These provide habitat for a variety of marine animals, including *P. guttatus*. The lobsters apparently remain in the interstices among the boulders during the day and crawl over the surfaces of the boulders at night. We have observed very few of these lobsters while diving during the day, but they are easily located at night. Schroeder & Starck (1964) photographed *P. guttatus* while diving at night on Alligator Reef east of Lower Matecumbe Key, Florida.

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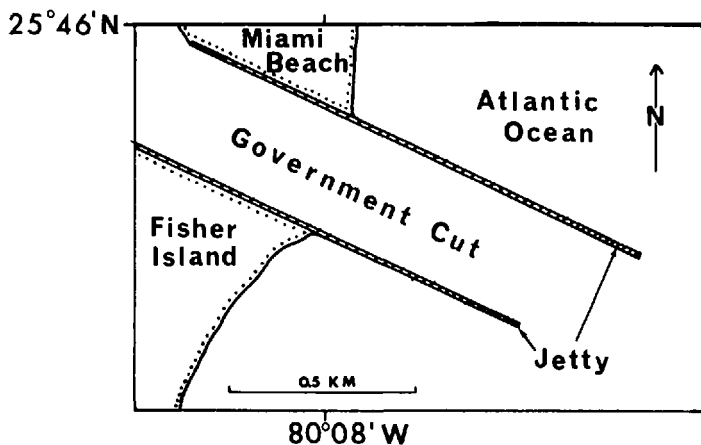


FIGURE 1. Jetties bordering Government Cut, the ship channel passing between Miami Beach and Fisher Island, Florida. (The jetty on the Miami Beach side of the channel is considered the north jetty, and that on the Fisher Island side, the south jetty.)

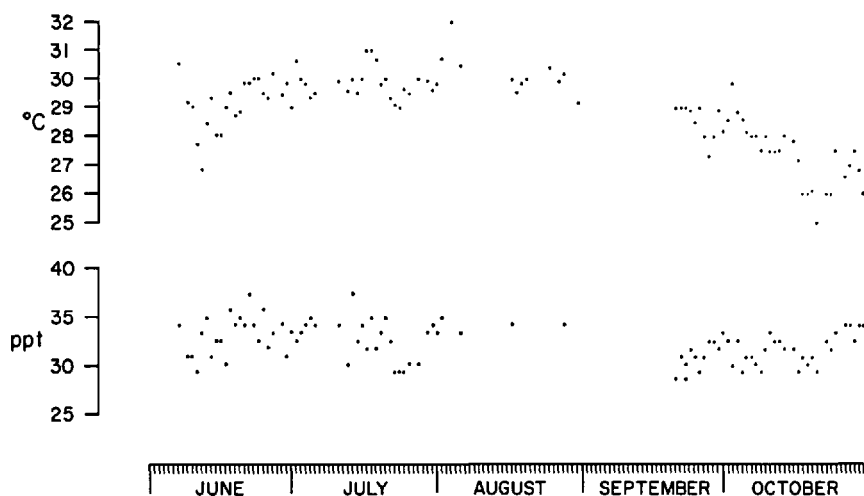


FIGURE 2. Temperature (°C) and salinity (parts per thousand, ppt) of surface water in Government Cut during June through October 1970. (The first day of each month is represented by the longer of the vertical marks on the horizontal axis.)

TABLE 1

DATE, LOCATION, COLLECTING METHOD, AND NUMBER OF MALES, NON-OVIGEROUS FEMALES, AND OVIGEROUS FEMALES OF *Panulirus guttatus* COLLECTED DURING JUNE THROUGH OCTOBER 1970

| Date*              | Location† | Collecting method‡ | No. males | No. females  |           | Total |
|--------------------|-----------|--------------------|-----------|--------------|-----------|-------|
|                    |           |                    |           | Nonovigerous | Ovigerous |       |
| June 11-12         | S         | A                  | 2         | 1            | 3         | 6     |
| 14                 | S         | A                  | 10        | 1            | 6         | 17    |
| 16                 | S         | A                  | 4         | 1            | 7         | 12    |
| 17                 | S         | A                  | 0         | 0            | 7         | 7     |
| 18                 | S         | A                  | 2         | 0            | 1         | 3     |
| 20                 | N&S       | B                  | 43        | 9            | 23        | 75    |
| 30                 | S         | A                  | 7         | 0            | 4         | 11    |
| June Subtotal      |           |                    | 68        | 12           | 51        | 131   |
| July 1             | S         | A                  | 2         | 2            | 0         | 4     |
| 2                  | S         | B                  | 69        | 29           | 10        | 108   |
| July Subtotal      |           |                    | 71        | 31           | 10        | 112   |
| August 1           | N         | B                  | 46        | 51           | 5         | 102   |
| 8-9                | N&S       | B                  | 64        | 44           | 17        | 125   |
| August Subtotal    |           |                    | 110       | 95           | 22        | 227   |
| September 3        | S         | B                  | 75        | 34           | 5         | 114   |
| 6                  | N         | B                  | 6         | 7            | 0         | 13    |
| September Subtotal |           |                    | 81        | 41           | 5         | 127   |
| October 1          | S         | B                  | 71        | 68           | 0         | 139   |
| 3                  | N         | B                  | 34        | 18           | 1         | 53    |
| 31                 | S         | B                  | 53        | 51           | 1         | 105   |
| October Subtotal   |           |                    | 158       | 137          | 2         | 297   |
| TOTAL              |           |                    | 488       | 316          | 90        | 894   |

\* When two dates are given, diving occurred before and after midnight on same night.

† S = south jetty; N = north jetty.

‡ A = dipnet; B = diving.

Two methods were used to collect *P. guttatus*. The most frequently used and most successful method was that of diving at night with snorkel, face mask, swim fins, and underwater light, and seizing the lobsters with gloved hand. Three divers could collect about 100 lobsters per hour by this method. Most diving was confined to the sides of the jetties within the ship channel, but some was done along the jetties outside the channel (Fig. 1). The other method, employed with minor success in June and early

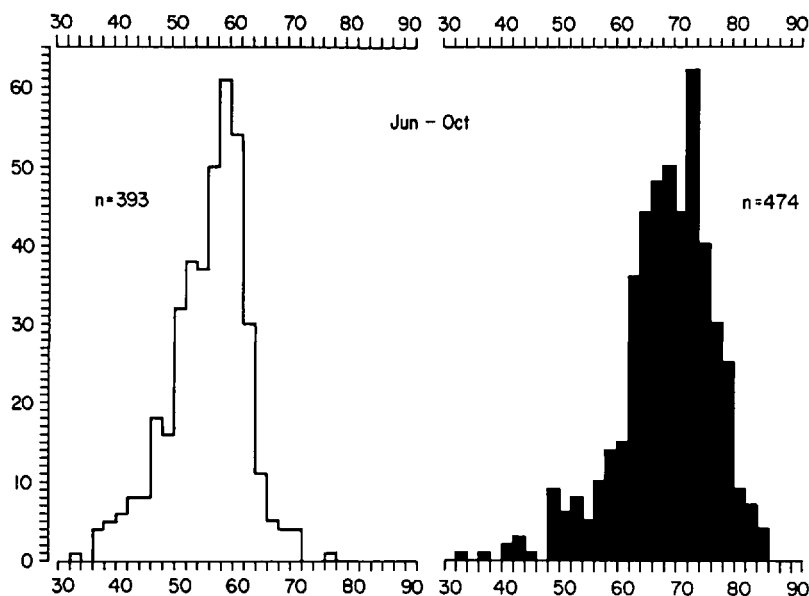


FIGURE 3. Frequency distributions of carapacial length in male (black histogram) and female (white histogram) specimens of *Panulirus guttatus* collected during June through October 1970. (Vertical scale represents number of specimens; horizontal scale represents carapacial length in mm divided into 2-mm classes, with the first digit of the numerical labels marking the lower limit of a class.)

July, was that of walking along the jetty on the south side of the channel at night, locating the lobsters with a light, and capturing them with a dipnet.

Water depth near the jetties ranges from 1.5 m westward to 6 m at the seaward ends of the jetties. The lobsters were more frequently found on the boulders within about 3 m of the water's surface than near, or on, the bottom. Lobsters were sought only during periods of flood and high slack tide, when the water was sufficiently clear.

Temperature and salinity of surface water were measured in Government Cut from the south jetty at intervals during the study (Fig. 2).

#### RESULTS AND DISCUSSION

Of the 894 specimens of *P. guttatus* collected during June through October 1970, 54.6 per cent were males and 45.4 per cent females (Table 1). Sutcliffe (1953) reported a ratio of males to females of 10 to 1 in his catches from Bermuda. Males and females were about equally represented in our June (48.1 per cent females), August (51.6 per cent females) and

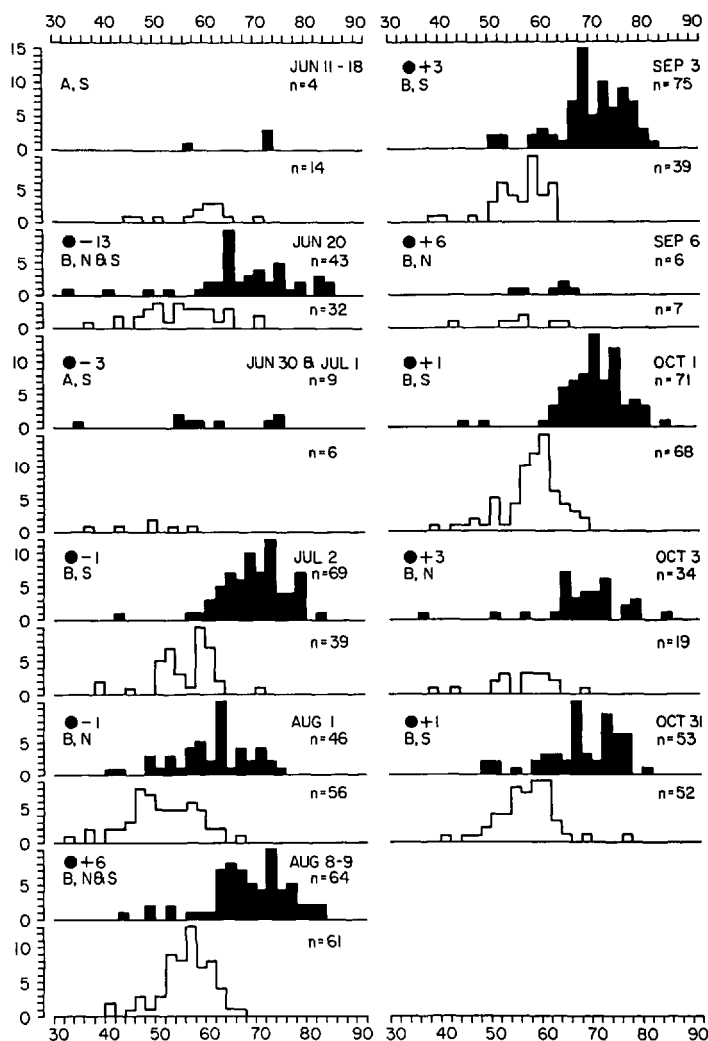


FIGURE 4. Frequency distributions of carapacial length in male (black histograms) and female (white histograms) specimens of *Panulirus guttatus*, by date of collection during June through October 1970. (Vertical scale represents number of specimens; horizontal scale represents carapacial length in mm, divided into 2-mm classes, with the first digit of the numerical labels marking the lower limit of a class; methods [A = dipnet, and B = diving] and location [S = south jetty, and N = north jetty] are shown for each collection, and number of days that the collection date preceded [-] or followed [+] the date of the new moon is given for the first date of each collection, except for the combined collections of June 11-18.)

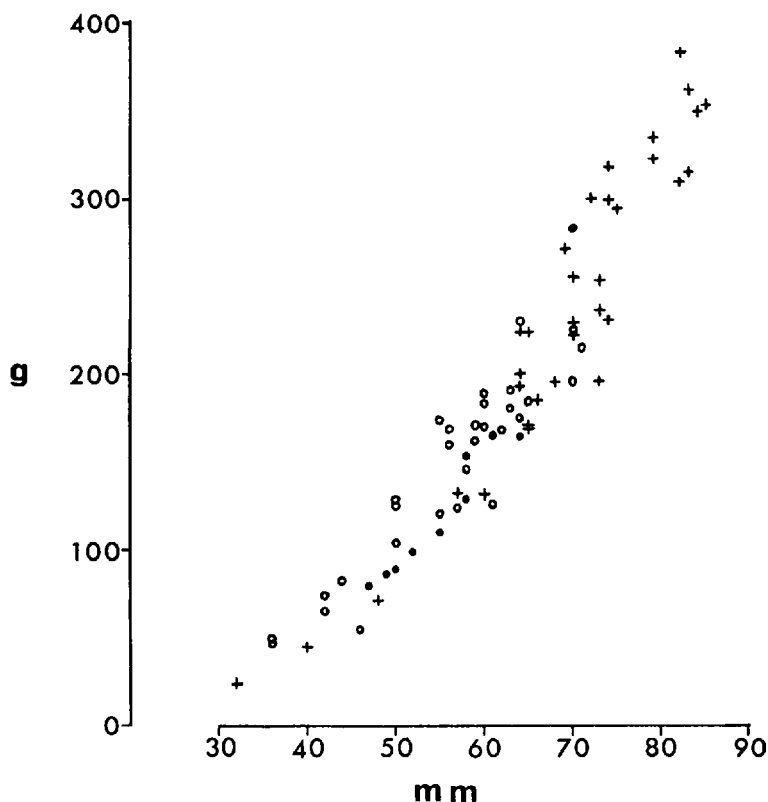


FIGURE 5. Relationships between weight (g) and carapacial length (mm) of male (+), nonovigerous female (●), and ovigerous female (○) specimens of *Panulirus guttatus* collected in June 1970.

October (46.8 per cent females) collections, but males outnumbered females in the July (36.6 per cent females) and September (36.2 per cent females) collections.

Of the females of *P. guttatus* caught, 81.0 per cent were ovigerous in June, 24.4 per cent in July, 18.8 per cent in August, 10.9 per cent in September, and 1.4 per cent in October (Table 1). The high proportion of ovigerous females in June and the decline in this proportion thereafter suggest that the peak spawning might occur in June or shortly before.

Measurements of carapacial length were made on 867 specimens (Fig. 3). Females of *P. guttatus* had a modal carapacial length of 58-59 mm, and that of males was 72-73 mm. Distribution of carapacial length did not vary greatly among collections, but there was some indication of an

TABLE 2

REGRESSION OF  $Y$  ( $= \log_{10}$  WEIGHT) ON  $X$  ( $= \log_{10}$  LENGTH) OF MALES, NONOVIGEROUS FEMALES, AND OVIGEROUS FEMALES OF *Panulirus guttatus* COLLECTED IN JUNE 1970\*

|                                               | Number | Range of carapacial length | Regression equation       | Standard deviation from regression |
|-----------------------------------------------|--------|----------------------------|---------------------------|------------------------------------|
| Males                                         | 33     | 32-85                      | $\hat{Y} = -2.90 + 2.85X$ | 0.06                               |
| Females                                       |        |                            |                           |                                    |
| nonovigerous                                  | 10     | 47-70                      | $\hat{Y} = -3.16 + 3.01X$ | 0.04                               |
| ovigerous                                     | 30     | 36-71                      | $\hat{Y} = -1.92 + 2.32X$ | 0.07                               |
| Females, non-ovigerous and ovigerous combined | 40     | 36-71                      | $\hat{Y} = -2.07 + 2.40X$ | 0.07                               |
| Sexes combined                                | 73     | 32-85                      | $\hat{Y} = -2.38 + 2.57X$ | 0.07                               |

\* Weight in g and carapacial length in mm.

increase in carapacial length of males and females from June to October (Fig. 4). The maximum carapacial length measured in males was 85 mm and that in females was 77 mm. The smallest ovigerous female had a carapacial length of 36 mm and weighed 45.8 g. Females of *P. guttatus* are smaller than males in Bermuda also (Sutcliffe, 1953).

Only 8.8 per cent of the 867 measured specimens of *P. guttatus* equalled or exceeded the minimum legal size limit of 3 inches (approximately 76 mm) in carapacial length now imposed to regulate the catch of *P. argus* in Florida, and all but one that did were males (Fig. 3). Were *P. guttatus* not recognized by Florida fishermen to be a species different from *P. argus* and therefore not protected by regulations, few *P. guttatus* would be included in the legal catch of spiny lobsters.

The relationship between weight and carapacial length of males, non-ovigerous females, and ovigerous females collected in June is shown in Figure 5. The slope of the regression of  $Y$  ( $= \log_{10}$  weight) on  $X$  ( $= \log_{10}$  length) of males was significantly ( $P < 0.05$ ) greater than that of females (nonovigerous and ovigerous combined, Table 2). Similar regressions for ovigerous and nonovigerous females are given separately in Table 2 for comparison.

At no time during the study was *P. argus* observed to be as abundant as *P. guttatus* along the jetties of Government Cut. Both species were observed on surfaces and crevices among the boulders at night, but *P. argus* was found more frequently near, or on, the bottom. Moore (1962) never

found the two species living together, for *P. guttatus* was always observed in an area not occupied by *P. argus*. Sutcliffe (1953) reported that the two species were taken in a ratio of 20 individuals of *P. argus* to 1 of *P. guttatus* on the outer reefs of Bermuda. He noted that *P. guttatus* was seldom taken outside the narrow band encompassing the outermost reefs or the relatively shallow water immediately adjacent, and unlike *P. argus*, it did not seem to undergo any noticeable migration.

#### ACKNOWLEDGMENTS

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#### SUMARIO

##### NOTAS SOBRE TAMAÑO, PROPORCIÓN SEXUAL Y DESOVE DE LA LANGOSTA *Panulirus guttatus* (LATREILLE), CERCA DE MIAMI BEACH, FLORIDA

Se colectaron langostas *Panulirus guttatus* (Latreille) durante la noche, a lo largo de los muelles que bordean el canal de barcos cerca de Miami Beach, Florida. Se presentan observaciones en la distribución del tamaño, proporción sexual y proporción de hembras ovígeras en la captura de langostas desde junio hasta octubre de 1970.

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